

**Department of Computer Science and Engineering
BRAC University**

Set A

Examination: Midterm
Duration: 1 hour 25 minutes

Semester: Spring 2025
Full Marks: 30

CSE 423: Computer Graphics

Name:	ID:	Section:
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[Answer the following questions. Understanding the questions is part of your examination. So, do not ask for any clarification of the questions.]

Question 1 [CO1]

- a. Matt wants to play Red Dead Redemption 2, requiring **144 FPS** on his pc, which uses a **32-bit color depth per pixel** for the frame buffer. A frame size in the buffer is **14,745,600 bytes**. Calculate the GPU's minimum rendering speed (pixels per millisecond) to achieve this FPS. [4]
- b. A robotic painter is positioned at an unknown location on a digital canvas where the line $4x + 9y + 128 = 0$ intersects the **x-axis**. The task for the painter is to draw a straight line from **this starting point** to the point **(-48, -17)**. Using the **Midpoint Line Drawing Algorithm**, compute the first 6 (including the starting point) pixels of the line segment the painter needs to use and **show** each step. [6]

Question 2 [CO3]

- a. Let's say the viewing region is a rectangle bounded by the points (0,0) and (100, 200). The bits in the region code are defined as follows:

Bit 3	Bit 2	Bit 1	Bit 0
Above	Below	Right	Left

- i. Leonard remembers that the number of iterations may vary in the **Cohen-Sutherland Line Clipping Algorithm**. Can you help him come up with an example line segment where the algorithm takes the **maximum number of iterations possible** in the given scenario? **State** the endpoints of your line segment and **draw** a rough illustration of your example. [4]
- ii. Sheldon looks at Leonard's example and says that he can reduce the number of iterations just by changing the **definition of the bits in the region code**. **State** the new region code definition (Sequence of Bits) that Sheldon has in mind. [1]

- b. A security camera monitors a restricted zone in a warehouse, recording movements only within (2, 2) to (8, 6). A drone follows a linear flight path from (3, 1) to (10, 7). Using the **Cyrus-Beck Line Clipping Algorithm**, compute the values of parameter t at which the drone enters and exits the detection area. Find the coordinates of the visible portion of the flight path. [5]

Question 3 [CO2]

- a. Why is scaling not considered a Euclidean transformation? Can all Euclidean transformations be written in the form of matrix multiplication? State your reasons. [2 + 2]
- b. A robot is positioned at (2, 0, 3) with the endpoint of its hand at (2, -2, 1). The robot wants to grab a fruit located at (12, 5, 13). To achieve this, it performs the following transformations sequentially on its hand's endpoint:
1. **Rotation**: The hand's endpoint is rotated 90° counter-clockwise around the X -axis, using the robot's position as the center of rotation.
 2. **Scaling**: The hand's endpoint is scaled uniformly by a factor of S .
 3. **Translation**: The robot itself is translated by (10, 0, 15), bringing the hand's endpoint to the fruit's position.
- Find the value of S that ensures the hand's endpoint aligns with the fruit (hand's endpoint will be the same as fruit's location). Show your calculations using composition of transformations. [6]

***** The End *****